

Take-Home Project – (APM)

Link to MVP prototype- [click here](#)

Name: Ayushmaan Singh

Email: ayushmaans2001@gmail.com

SPECIFYING THE PROBLEM

Motor insurance claims in India are slow, inconsistent, and costly due to legacy systems and manual processes, creating poor customer experience and fraud risk—requiring an agentic AI, automated workflow that improves speed, accuracy, and control.

WHY DOES THE PROBLEM EXIST

Operational Inefficiencies:

- Heavy manual work: Intake, photo triage, evidence gathering, and adjudication are manual and repetitive- *reactive and biased*
- Slow evidence capture: Claimants take low-quality photos; adjusters must request re-shoots or physical appraisals.
- Fragmented systems: legacy policy/claim systems, email, spreadsheets, and 3rd-party estimators don't talk well.

Decision Quality & Risk Issues:

- Lack of unified knowledge leads to variable settlements
- Fraud exposure- limited automated fraud signals cause leakage.



WHY SHOULD WE SOLVE THIS PROBLEM

Better Customer Experience

- Faster settlements $\Rightarrow$ higher NPS and retention.
- Enhanced trust and transparency with real-time status.
- Consistent, evidence-based decisions $\Rightarrow$ fewer disputes.

Efficiency & Cost Impact

- Unified workflow $\Rightarrow$ fewer manual touchpoints and errors.
- Better utilization of adjusters $\Rightarrow$ focus on higher-value tasks.
- Scalable operations $\Rightarrow$ handle claim surges
- Lower operational cost $\Rightarrow$ improved combined ratio and margin
- Quicker fraud detection $\Rightarrow$ reduced unnecessary payouts.



POTENTIAL IMPACT ON SOLVING THE PROBLEM – FOR

Core Assumptions

- Annual motor insurance claims filed in India: **10 M**^[1]
- Average claim payout: **₹30K**^[2]
- Operational processing cost per claim: **₹2K**^[1]



Baseline Annual Economics

- Total Claim Payouts: **30K Cr**
(10M claims $\times$ ₹30K)
- Total Operational Cost: **2K Cr**
(10M claims $\times$ ₹2K)
- Total Annual Claims Cost: **32K Cr**
(30K Cr + 2K Cr)



AI-Driven Improvement

- Payout optimization (better adjudication + repair estimation + fraud reduction): **5% savings**
- Operational automation (intake, triage, evidence validation): **40% savings**



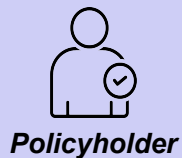
Resulting Annual Savings

- Savings from Payout Optimization (5%)
(5% $\times$ ₹30K Cr) = **₹1,500 crore annually**
- Savings from Operational Automation (40%)
(40% $\times$ ₹2K Cr) = **₹800 crore annually**
- Total Conservative Annual Savings:
₹2,300 Cr



UNDERSTANDING TARGET USERS – PAIN POINTS

What makes life difficult?



Policyholder

- **Confusing FNOL** channels and long **hold times**.
- Inability to submit **good evidence** (bad photos, missing info)
- **Long time** to first payment/repair authorization.



Claim Adjuster

- **Manual** evidence collection and chasing claimants/shops.
- Slow **damage estimation** process; follow-ups with appraisers.
- **Hard to prioritize** high-risk vs low-value closeable claims.



SIU

- High **false-positive** rates when triaging suspicious claims.
- **Delays** in receiving consolidated evidence and timelines.
- **Manual** report preparation and **poor escalation** decisions.



Intake Specialist

- Manual, **repetitive data capture** across multiple systems
- **Poor tools** for quickly validating policy at intake
- No influence over **evidence quality** (photo advice missing)



Appraiser/Estimator

- Poor quality images causing **re-inspections**.
- **Fragmented** cost databases for parts, labour
- **Slow turn-around** for estimates and approvals.



Claim Manager

- Lack of **real-time KPIs**; stale dashboards.
- Hard to optimize **workforce allocation** and SLA adherence
- Difficult to measure **adjuster effectiveness**/bottlenecks

IDENTIFYING PRIORITY USER

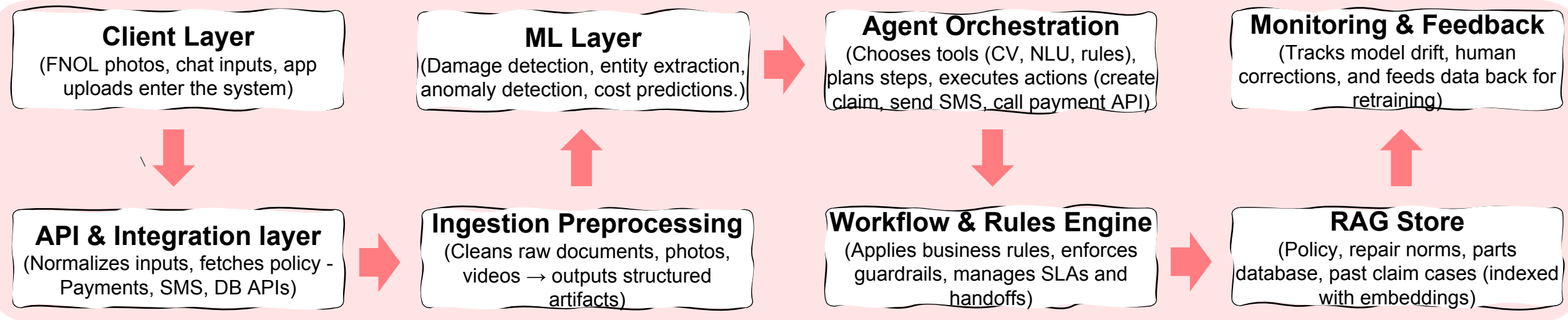
Target User	Business Impact	Frequency & effort	Feasibility	Ease of adoption
Policyholder	Medium	Medium	High	Medium
Intake Specialist	Medium	High	High	High
Claims Adjuster	High	High	High	High
Appraiser/Estimator	High	Medium	Medium	Medium
SIU	High (For fraud cases)	Low	Medium	Medium
Claims Manager	Medium	Low	Medium	Medium
Reasoning (for HIGH's)	They largely influence payout	They touch most claims	Most work is manual/low risk (easy to automate)	They will get immediate productivity gains

SOLUTION: KAI (Keep Adjuster-Informed) - Agentic Claims

KAI is an agentic assistant that automates intake, evidence capture, triage, damage/fraud analysis, and low-risk settlements—while keeping adjusters in control for exceptions.

Features	Core Function	Agentic Autonomy	Business value
Agentic FNOL	Intake via chat; guides the claimant to click right photos, ask clarifying questions	- Uses image-quality check; gives feedback - Uploads structured evidence to claim records	Cuts re-inspections, boosts estimate accuracy, shortens FNOL-to-triage time
Automated Damage Triage	- Predicts severity scale from claimant data - Preliminary estimate using regional rates	Vision model detects parts, estimates repair complexity; maps to price items using catalog	Faster approvals for low-risk claims, reduce appraiser load.
Risk & Fraud Scoring	Collects signals- claimant history, image tampering, geolocation to build risk score	Generates a human-readable explainable summary; If high risk, auto-creates SIU case	Reduce leakage and false positives, faster SIU investigations.
Smart adjuster Assignment	Assigns claims to the right adjuster based on complexity, location, workload, skills	- Monitors workloads and rebalances in real time - Can auto-schedule 3rd-party appraisers	Better resource utilization and SLA adherence.
Adjuster Workspace	Single pane showing all evidence, a brief claim summary, and a suggested decision	- Can execute low-risk payouts under guardrails. - Give audit trail and RAG (rationale + evidence)	Faster claim resolutions due to pre-assembled evidence and decisions.

TECHNICAL DESIGN



MVP PRIORITIZATION – Scope, Features, Rationale

Feature Parameter	Agentic FNOL	Automated Damage Triage	Risk & Fraud Scoring	Smart adjuster Assignment	Adjuster Workspace
Time-to-market	▲▲▲▲	▲▲	▲	▲▲▲	▲
Business value	▲▲▲▲	▲▲▲▲	▲▲▲	▲	▲▲▲
Feasibility	▲▲▲	▲▲▲	▲▲	▲▲▲	▲▲
Risk / Compliance	▲▲▲	▲▲▲	▲▲	▲▲▲▲	▲▲▲
Measurability	▲▲▲▲	▲▲▲	▲▲▲	▲▲▲	▲▲
Net Result	MVP	MVP	Phase-2	Phase-1	Phase-2

QUALITATIVE REASONING – [Link to prototype](#)

Agentic FNOL

- 1. Fast to build:** LLM conversation + photo checks
- 2. High impact:** Fixes intake quality → fewer re-inspections, cleaner evidence
- 3. Low risk:** Only collects data; fully auditable; no financial decisions.
- 4. Easy to measure:** Completion rate, photo quality, missing fields → measurable day 1.

Automated Damage Triage

- 1. Medium-fast build:** Pre-trained CV + rules-based pricing → quick prototype.
- 2. Huge value:** Instant severity tiering reduces adjuster load, speeds low-risk claims.
- 3. Feasible:** Mature models + safe fallback to adjuster when confidence low.
- 4. Low risk:** Only suggests next step; no automated payout

Risk & Fraud Scoring

- 1. Slow to build:** Needs historical fraud data + integrations.
- 2. Higher risk:** Wrong signals → legal + customer impact.
- 3. Hard to measure:** Fraud metrics need months of data.
- 4. Complex feasibility:** Requires explainability + monitoring.

Smart Adjuster Assignment

- 1. Medium complexity:** Needs roster, skills, workload integrations.
- 2. Moderate impact:** Improves SLAs but not as big as FNOL/triage.
- 3. Low risk:** Internal workflow, no financial payout
- 4. Moderate measurability:** Needs volume to show SLA

Adjuster Workspace

- 1. High build effort:** Full UI + backend + evidence stitching.
- 2. Dependent:** Needs FNOL + triage + fraud outputs to be effective.
- 3. Moderate feasibility:** Heavy SOR integration.
- 4. Measured only later:** Gains visible only after adjuster adoption.

Link to MVP Prototype

[Click here](#)